

Ashwin Kumar V

APPLIED AI / ML ENGINEER | UI / UX DESIGNER | DEVELOPER | DESIGN LEAD - TENSOR CLUB

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ABOUT ME

Driven AIML & Software Development enthusiast focused on bridging the gap between innovative code and practical applications. Experienced in training neural networks, designing intelligent software solutions, and exploring cutting-edge AI technologies. Eager to contribute technical expertise and a collaborative mindset to a forward-thinking development team.

EDUCATION

B.E in Artificial Intelligence & Machine Learning	MVJ College of Engineering , Bengaluru	2023 - 2027	8.5 CGPA
Pre - University : PCME'S	ST Joseph's Indian Composite PU College	2021 - 2023	85%
Schooling	Narayana E-Techno School	2018 - 2021	81%

TECHNICAL SKILLS

- **Programming Languages:** Python, C , C++ , SQL
- **AI & ML:** Machine Learning, Deep Learning , Computer Vision, Natural Language Processing (NLP) , RAG , Langchain
- **Libraries & Frameworks:** OpenCV, TensorFlow, Keras, PyTorch, NumPy, Pandas, Scikit-learn
- **Tools & Developer Platforms:** Git/GitHub, Overleaf, LaTeX (IEEE Formatting)
- **Web & Interface Concepts:** UI/UX Concepts, 3D/Interactive Web Principles , Figma , Webflow , Dribble , Photoshop

EXPERIENCE

Design Lead - Tensor Club , MVJCE | BENGALURU

- Led UI/UX design and visual branding for tech events, hackathons, and software projects.
- Collaborated with developers to translate complex AI/ML concepts into intuitive user interfaces.
- Designed standardized technical templates using LaTeX and IEEE formatting for club research.
- Managed a team of student designers to establish a cohesive digital identity for the club.
- Created high-impact promotional assets that boosted event registration and student engagement.
- Bridged the gap between design and tech by integrating interactive workflows into web projects.

Technical Projects & Execution

- Web Development: Designed responsive front-ends with interactive UI/UX elements, merging aesthetic layouts with clean code.
- Machine Learning: Developed deep learning models (CNN/NLP) and computer vision pipelines for software-based automation.
- Core Technology: Managed isolated Python virtual environments and authored structured IEEE-standard LaTeX documentation.

PROJECTS

RAG Document Retrieval Pipeline

- Built a full-stack RAG pipeline using Python and LangChain to extract data from PDFs and enable natural language querying.
- Integrated FAISS vector database for efficient semantic search, text chunking, and high-accuracy document retrieval.
- Utilized HuggingFace Embeddings and LLMs to generate precise, context-aware answers from processed text.

Local AI Agent with Offline RAG Architecture

- Developed a 100% private, offline AI agent using Python, LangChain, and Ollama to process and query local PDFs, CSVs, and text files without external API or cloud dependencies.
- Integrated Llama 3.2 locally via Ollama for natural language understanding, using LangChain to manage prompt chaining and retrieval logic.
- Built a high-efficiency vector storage layer utilizing ChromaDB and mxbai-embed-large embeddings to execute fast, cost-efficient semantic searches.
- Designed a secure, privacy-focused RAG workflow that extracts real-time facts from local data directories to generate context-aware, precise responses.

Conversational Voice AI Pipeline

- Engineered a real-time conversational Voice AI Assistant using Python and Vapi AI to automate outbound customer service calling and dynamic order tracking.
- Integrated a backend database to instantly verify customer-provided order numbers, programmatically managing call flows based on data validation.
- Deployed a secure local server gateway using Ngrok to expose endpoints for seamless, low-latency, real-time voice communication and API synchronization.
- Implemented fallback handling and call state logic to gracefully manage invalid entries, optimizing user experience and conversational flow.

ADDITIONAL PROJECTS : ML PROJECTS (SPAM MODEL, DISEASE PREDICTOR,AMBULANCE DETECTION MODEL)

ADDITIONAL DETAILS

Languages : English , Hindi , Kannada, Spanish: Basic (In Progress)

Certifications : DSA Using Python (UDEMY) , Building 3D models.

Hobbies : Sports , Sketching , Video games

Achievements:

- Second runner up in **Ideathon** (MVJCE)
- Runner up in Treasure Hunt , Cultural program